

curriculum vitae

Versatile R&D Engineer with extensive experience in computational mechanics, finite element analysis, digital image correlation, and physics-based surrogate model development. Specializes in algorithm design across Matlab, Python, C++, and Linux (bash) for complex engineering workflows. Combines deep computational mechanics expertise with hands-on experimental work. Effective communicator who bridges detailed scientific analysis with real-world application needs.

Personalia

Surname:	Ruybalid
Given names	Andre, Paul
Degrees:	Ph.D. MSc. BSc.
City:	Los Alamos, New Mexico, U.S.A.
Phone number:	+1 719 210 1746 (personal mobile)
E-mail:	andreruybalid@gmail.com
Date of birth:	26 October 1987
Place of Birth:	Farmington, New Mexico, U.S.A.
Nationality:	American/Dutch

Professional experience

2023 - Present

R&D Engineer, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Developed multiple surrogate models and custom Python-based tools to bridge polycrystalline mechanics to engineering-scale predictions, demonstrating algorithm design across Python, C++, and Linux (bash) for complex engineering workflows. Extensive experience building automation around complex engineering software and turning research into production-ready systems.
- ❖ Built and maintained material models for nuclear fuel performance analysis in MOOSE/BISON, including code contributions, verification, validation, and uncertainty quantification. Demonstrated ability to design deterministic, reliable automation around complex simulation tools.
- ❖ Collaborated with external industrial partners and national lab colleagues to deliver pragmatic modeling solutions; led technical milestone reports, peer-reviewed papers, and presented at scientific conferences.
- ❖ Reached the R&D 100 Award Finals twice with the surrogate modeling solutions.

2021 – 2023

Postdoctoral Researcher, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

Developed the surrogate model ("LAROMance") for polycrystalline nuclear reactor materials by designing large-scale simulation pipelines and curating datasets from thousands of low length-scale simulations. Involved a long-term research project with evolving requirements to build reusable, scalable modeling capabilities.

2019 – 2020

Design Engineer, ASML

DUV Def & Metrology group of Defectivity Department | Eindhoven, The Netherlands

Developed and validated advanced metrology systems for high-precision nanoscale defect detection on critical DUV lithography reticles, performing specialized cleanroom testing and data analysis under tight deadlines in collaboration with cross-functional and industrial teams.

2013 – 2019

PhD Researcher, Eindhoven University of Technology

Computational and Experimental Mechanics | Prof. Dr. Marc Geers & Dr. Johan Hoefnagels.

Philips Research, Materials Innovation Institute | Eindhoven, The Netherlands

Developed a numerical-experimental workflow combining digital image correlation with finite element modeling to measure adhesion properties, utilizing a custom micromechanical test rig with automated feedback control hardware and software

- algorithm design for precise mechanical force measurement. Resulted in five journal publications.
- 2017 – 2020** **Account manager & Technical recruiter**
Research-Square & EuFlex | Eindhoven, The Netherlands
 Acquired and managed client accounts, and conducted technical recruitment of highly specialized engineering and PhD-level candidates for science and technology companies.
- 2018 – 2019** **Lab technologist, Multi-Scale Laboratory**
Eindhoven University of Technology | Eindhoven, The Netherlands
Computational and Experimental Mechanics
 Managed an experimental mechanics laboratory, overseeing high-tech equipment including scanning electron microscopes and micro-mechanical test systems, while providing technical guidance to researchers and students and supporting experimental design in a high-demand environment.
- 2009 – 2013** **Tutor of students**
Eindhoven University of Technology, Mechanical Engineering Department | Freelance
 Mentored bachelor student teams on engineering case studies, supporting balanced group dynamics, resolving team issues, and delivering constructive feedback.

Education

Eindhoven University of Technology, the Netherlands
Department of Mechanical Engineering
Division: Computational and Experimental Mechanics
Research Group: Mechanics of Materials

- 2013 – 2019** **PhD.-research project**
Title: "Identification of multi-layer interface systems: a full-field, micro-mechanical approach."
- 2013 – 2019** **Graduate School of Engineering Mechanics**
Advanced Mechanics Courses
- 2010 – 2013** **MSc. Mechanical Engineering**
Minor: Biomedical Engineering
- 2006 – 2010** **BSc. Mechanical Engineering**
- 2000 – 2006** **Research-oriented secondary education (VWO)**
St.-Janscollege, Hoensbroek, The Netherlands

Selected journal publications (peer-reviewed)

P. Robbe, A.P. Ruybalid, A. Hegde, C. Bonneville, H.N. Najm, L. Capolungo and C. Safta . A comparison of surrogate constitutive models for viscoplastic creep simulation of HT-9 steel. *Computational Materials Science*, 262, 114367 (2026). <https://doi.org/10.1016/j.commatsci.2025.114367>.

A.P. Ruybalid, A. Tallman, W. Wen, C. Matthews, L. Capolungo. Data-Driven Surrogate Modeling with Microstructure-Sensitivity of Viscoplastic Creep in Grade 91 Steel. *Integrating Materials and Manufacturing Innovation* 13, 895–914 (2024). <https://doi.org/10.1007/s40192-024-00377-z>

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers and J.P.M. Hoefnagels. Full-field identification of mixed-mode adhesion properties in a flexible, multi-layer microelectronic material system. *Engineering Fracture Mechanics*, 226, pp. 106879 (2020), <https://doi.org/10.1016/j.engfracmech.2020.106879>.

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers, J.P.M. Hoefnagels. An *In-Situ*, Micro-Mechanical Setup with Accurate, Tri-Axial, Piezoelectric Force Sensing and Positioning. *Exp Mech* 60, 713–725 (2020), <https://doi.org/10.1007/s11340-020-00583-8>.

A.P. Ruybalid, J.P.M. Hoefnagels, O. van der Sluis, M.P.F.H.L. van Maris and M.G.D. Geers. Mixed-mode cohesive zone parameters from integrated digital image correlation on micrographs only. *International Journal of Solids and Structures*, 156-157, pp. 179-193 (2019), <https://doi.org/10.1016/j.ijsolstr.2018.08.010>.